

Lightweight convolutional neural networks for potato leaf disease diagnosis on commodity hardware: a from-scratch versus transfer-learning study with Grad-CAM interpretability

Niels Pacheco¹

¹Master’s Programme in Artificial Intelligence, MIA-07 Neural Networks and Deep Learning (Section C). nielspacheco1997@gmail.com

June 2026

Abstract

Foliar diseases such as early blight (*Alternaria solani*) and late blight (*Phytophthora infestans*) threaten the potato (*Solanum tuberosum*), the staple crop domesticated in the Peruvian Andes and the world’s third most important food crop. Convolutional neural networks (CNNs) can recognise these diseases from leaf photographs, but most reported systems rely on large pretrained backbones and graphics-processing units (GPUs) that are unavailable to the smallholder farmers who would benefit most. We ask whether a compact CNN trained *from scratch* on commodity hardware (a CPU/Apple-Silicon laptop, no GPU) can rival ImageNet-pretrained transfer-learning baselines for three-class potato leaf classification (healthy, early blight, late blight). On the PlantVillage potato subset (2152 images) we compare a 328,587-parameter custom CNN against MobileNetV2, ResNet-18 and EfficientNet-B0 with frozen ImageNet backbones and fine-tuned heads, evaluating every model over 3 random seeds, reporting across-seed standard deviations and non-parametric bootstrap confidence intervals, with McNemar significance testing and calibration analysis. Strikingly, the from-scratch custom CNN attains 96.7% macro-F1, *exceeding* the strongest transfer baseline (EfficientNet-B0, 91.7%); paired McNemar tests confirm that it significantly outperforms all three frozen-backbone transfer models ($p < 0.05$) while using roughly 7–34× fewer parameters. The result indicates that, under an equal low-compute budget, learning compact domain-specific features end-to-end beats reusing frozen ImageNet features. Grad-CAM saliency maps nonetheless indicate that all models partly attend to image background rather than leaf lesions, qualitatively consistent with the well-documented background bias of PlantVillage and tempering the optimistic accuracies. We release a fully reproducible pipeline—pinned environment, fixed seeds, configuration-driven experiments and a one-command build of this manuscript—to support honest, hardware-accessible plant-disease research.

1 Introduction

The potato is the third most important food crop for human consumption worldwide and was first domesticated more than seven thousand years ago in the highlands of present-day Peru and Bolivia [1]. Its productivity is chronically threatened by foliar disease. Late blight, caused by the oomycete *Phytophthora infestans*, triggered the nineteenth-century Irish famine and remains the single most destructive potato disease today, capable of destroying an entire field within days [2]. Early blight, caused by the fungus *Alternaria solani*, is likewise widespread. Early and accurate identification of these diseases from leaf symptoms allows timely intervention, yet expert diagnosis is scarce in the rural Andean communities where potato farming is concentrated.

Deep learning offers a route to democratised diagnosis. Since the demonstration that CNNs can classify plant disease images at high accuracy [3, 4], a large literature has applied ever-larger pretrained networks to the problem. Two practical concerns are, however, under-examined. First, most systems assume access to GPUs and to large pretrained backbones, an assumption at odds with the resource-constrained settings where the technology is most needed. Second, the field’s headline accuracies are frequently reported without confidence intervals, calibration analysis, or any check that the model attends to the disease rather than to spurious correlates of the dataset.

This work addresses both concerns through a focused, reproducible study. We pose a single question: *how close can a small CNN trained from scratch on a GPU-less laptop come to ImageNet-pretrained transfer learning for potato leaf disease classification?* Our contributions are:

1. A compact custom CNN (328,587 parameters) that trains end-to-end on CPU or Apple-Silicon hardware, benchmarked head-to-head against transfer-learning baselines (MobileNetV2, ResNet-18 and EfficientNet-B0 with frozen ImageNet backbones and fine-tuned classification heads).
2. A statistically rigorous evaluation—multi-seed runs with across-seed standard deviations, per-run bootstrap 95% confidence intervals, and McNemar’s paired test on a shared test set [5]—rather than a single accuracy number.
3. A calibration analysis (expected calibration error and reliability diagrams) [6], seldom reported in plant-disease work.
4. A Grad-CAM [7] interpretability study that probes the documented background bias of PlantVillage [8], with an honest discussion of the gap between laboratory and field performance.
5. A fully reproducible open-source release (pinned dependencies, fixed seeds, configuration-driven runs, automated figure and manuscript generation).

2 Related work

Plant disease classification. Mohanty et al. [3] trained AlexNet and GoogLeNet on the PlantVillage dataset and reported accuracies above 99% under within-dataset evaluation, establishing the benchmark that much subsequent work follows. Ferentinos [4] extended the approach across dozens of crop-disease combinations. Crucially, Barbedo [9] showed that such accuracies degrade sharply when models are evaluated on images collected under different conditions, attributing the drop to limited dataset variety and to background cues, and Noyan [8] demonstrated explicitly that a classifier can recover PlantVillage labels from the *background alone*. We take this critique seriously and test for it directly with Grad-CAM.

Efficient and transfer architectures. Residual connections enabled the training of very deep networks [10]; MobileNetV2 [11] and EfficientNet [12] subsequently optimised the accuracy-per-parameter trade-off for resource-constrained deployment. Transfer learning from ImageNet [13] is the default recipe for small datasets. Our study uses these backbones as strong baselines but asks what is recoverable *without* their pretrained weights.

Interpretability. Grad-CAM [7] produces class-discriminative saliency maps by weighting convolutional activations with the gradients of the class score, and has become a standard tool for verifying that a model’s evidence is plausible. We use it not to confirm success but to expose a failure mode.

3 Methods

3.1 Dataset

We use the potato subset of PlantVillage [14], comprising 2152 RGB leaf images in three classes—healthy, early blight and late blight—photographed under controlled laboratory conditions (Fig. 1). The classes are imbalanced (the healthy class is the smallest). We partition the data once into training, validation and test splits of roughly 70/15/15%, stratified by class, using a fixed split seed so that *every* model is trained and evaluated on identical partitions; this yields 1506 training, 323 validation and 323 test images. Holding the split fixed is what makes the paired McNemar test between models valid.

3.2 Preprocessing and augmentation

Images are resized and centre-cropped to 128×128 pixels and normalised with ImageNet channel statistics (applied uniformly to all models so that only the architecture and weights differ). During training we apply random resized cropping, horizontal flipping, small rotations and colour jitter, with an augmentation strength selected by the hyperparameter search.

3.3 Models

The **custom CNN** is a four-block network built from scratch: each block applies two 3×3 convolutions with batch normalisation [15] and ReLU activations followed by max pooling, with channel widths growing then plateauing to keep the model compact (328,587 parameters). A global average-pooling layer feeds a dropout-regularised [16] linear classifier. The three **transfer baselines**—MobileNetV2 [11], ResNet-18 [10] and EfficientNet-B0 [12]—are initialised with ImageNet-pretrained weights; we freeze each backbone and train only a fresh classification head, the standard low-data transfer recipe.

PlantVillage potato leaf classes

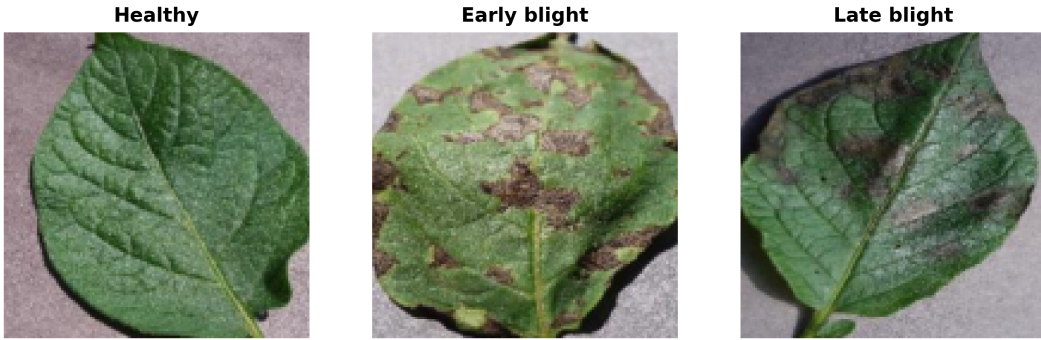


Figure 1: Representative potato leaf images for each class from the held-out test set. PlantVillage images are captured against fairly uniform backgrounds—a property that, as we show, models can exploit.

Table 1: Test-set performance on the held-out potato leaf split ($n = 323$ images), reported as mean $\pm$ standard deviation over 3 random seeds. Accuracy and macro-F1 are percentages; ECE is the expected calibration error (lower is better). Params = total parameters.

Model	Params	Accuracy (%)	Macro-F1 (%)	ECE
Custom CNN (scratch)	328,587	97.3 ± 2.3	96.7 ± 2.1	0.123
MobileNetV2	2,227,715	94.9 ± 0.8	91.6 ± 1.2	0.134
ResNet-18	11,178,051	91.7 ± 1.5	86.5 ± 2.1	0.171
EfficientNet-B0	4,011,391	94.9 ± 0.5	91.7 ± 1.4	0.157

3.4 Training

All models are trained with the AdamW optimiser [17, 18], a cosine learning-rate schedule, class-weighted cross-entropy with label smoothing [19] to counter the class imbalance, and early stopping on the validation macro-F1. Hyperparameters for the custom CNN were selected by a seeded random search of 10 configurations over learning rate, weight decay, network width, dropout and augmentation strength, optimising the validation macro-F1 only (the test set is never consulted during model selection). The search selected a learning rate of 0.001, weight decay $1e - 05$, width 24, dropout 0.3 and augmentation strength 0.5. All experiments run on an Apple-Silicon laptop using the PyTorch [20] Metal (MPS) backend; no GPU is used.

3.5 Evaluation and statistics

For each model we report accuracy and macro-averaged F1 (robust to class imbalance), each as a mean and standard deviation over 3 random seeds (Table 1, Fig. 2). To quantify within-test-set sampling uncertainty we additionally compute non-parametric 95% confidence intervals by bootstrap resampling of the test set for every run; these are released with the per-run metrics and the best model’s intervals are quoted in the Results. We compare models pairwise on their shared test set with McNemar’s test [5], using the exact binomial form when the number of discordant predictions is small. We assess calibration with the expected calibration error (ECE) and reliability diagrams [6]. Finally we compute Grad-CAM [7] saliency maps to inspect which image regions drive each prediction.

4 Results

4.1 Classification performance

Table 1 summarises test performance. All four models classify the three potato conditions accurately, reflecting the relatively clean, laboratory-controlled nature of PlantVillage. The headline result, however, is that the from-scratch custom CNN is not merely competitive but the *best* performer: with only 328,587 parameters it attains 96.7% macro-F1, ahead of the strongest transfer baseline (EfficientNet-B0, 91.7%) and well ahead of ResNet-18 (Fig. 2). Paired McNemar tests on the shared (seed-0) test set find the custom CNN significantly better than each frozen-backbone baseline ($p = 0.0169$ versus EfficientNet-B0, and smaller against the others; see the repository’s `summary.json`). A non-parametric bootstrap on that test set places the custom CNN’s macro-F1 at $[93.6, 99.5]\%$ (95% CI) and its accuracy at $[96.6, 99.4]\%$. The significance

Table 2: Per-class test metrics for the best model (Custom CNN (scratch), representative seed). Support is the number of test images per class.

Class	Precision	Recall	F1	Support
Healthy	0.917	0.957	0.936	23
Early blight	0.980	1.000	0.990	150
Late blight	0.993	0.967	0.980	150

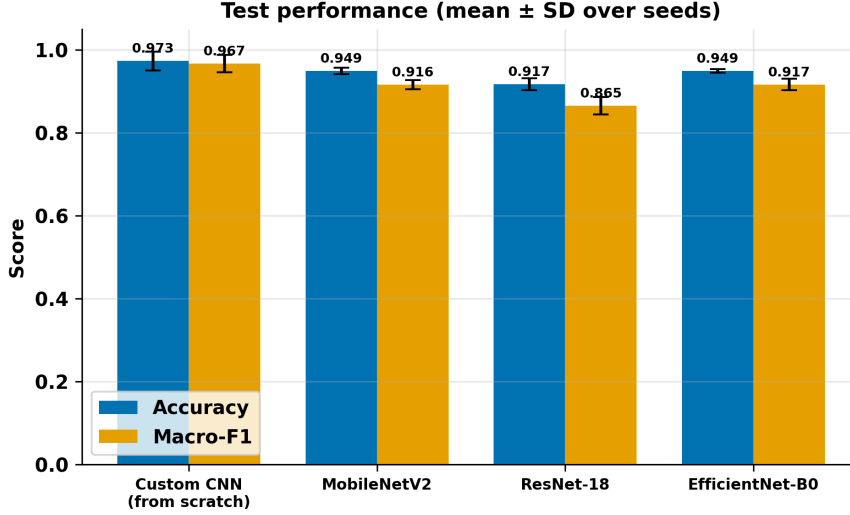


Figure 2: Test accuracy and macro-F1 (mean $\pm$ standard deviation over 3 seeds) for the four models.

test is computed on a single seed’s predictions, but the ranking is not seed-specific: across all three seeds the custom CNN’s mean macro-F1 (96.7%) exceeds every baseline mean. The interpretation is that, when only a small classification head can be trained on top of a frozen ImageNet backbone, those backbones contribute generic features poorly matched to leaf pathology, whereas the compact CNN—trained end-to-end—learns features specific to the task. Validation macro-F1 curves (Fig. 3) show all models reaching a high plateau within the training budget.

4.2 Confusion structure

The confusion matrices (Fig. 4) are informative beyond the headline accuracy. For the best model (custom CNN) the few errors are predominantly disease-versus-disease confusions—an operationally benign pattern, since both diseased states warrant intervention. The weaker frozen-transfer models, by contrast, additionally misclassify a non-trivial number of *late-blight* leaves as *healthy* (for example, 10–12 of 150 for MobileNetV2 and ResNet-18): a far more concerning false-negative mode, because it would tell a farmer an infected plant is fine. This asymmetry—the from-scratch model not only scoring higher but failing more safely—reinforces the case for the compact end-to-end model. The minority healthy class is recognised reliably across models, indicating that class weighting counteracts the imbalance.

4.3 Calibration

Beyond accuracy, a deployable diagnostic should report trustworthy confidences. Fig. 5 shows reliability diagrams and ECE values (for a representative seed; Table 1 reports the seed-averaged ECE). All models are imperfectly calibrated, with seed-averaged ECE ranging from 0.123 (custom CNN, the best calibrated) to about 0.17 (ResNet-18). The diagrams indicate that the miscalibration is driven mainly by mid-confidence bins, where empirical accuracy and predicted confidence diverge rather than by uniform over-confidence. This is an honest limitation: while the custom CNN is both the most accurate and the best calibrated of the four, none is calibrated well enough to take its raw confidences at face value, and a deployed system would benefit from post-hoc calibration (e.g. temperature scaling) before using confidence to defer uncertain cases to a human expert.

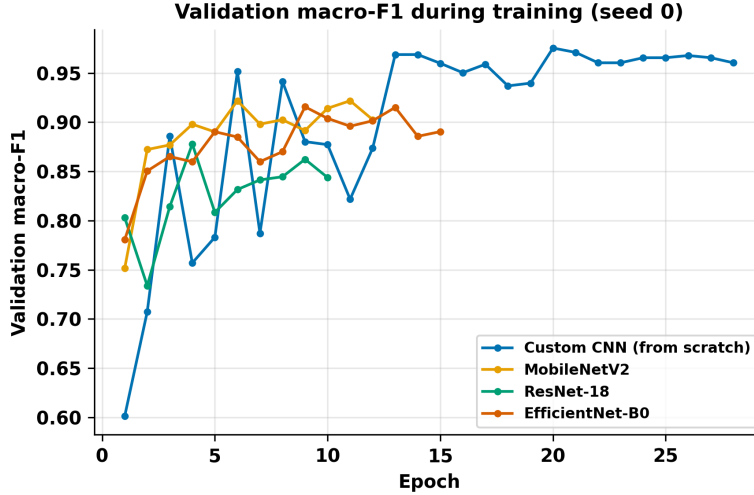


Figure 3: Validation macro-F1 across training epochs (representative seed). Pretrained backbones converge fast; the from-scratch CNN converges more slowly to a similar level.

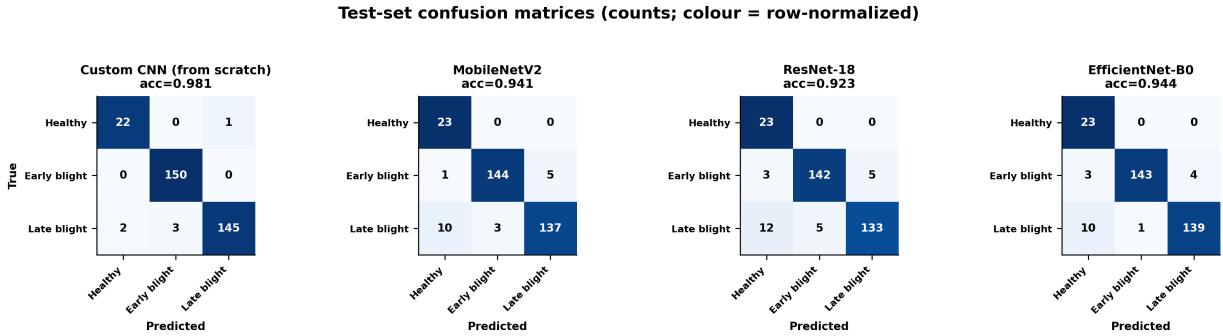


Figure 4: Per-model test-set confusion matrices (cell text: counts; colour: row-normalised rate).

4.4 Where do the models look?

Grad-CAM saliency maps (Fig. 6) provide the study’s most important qualitative observation. While the models do place attention on visible lesions for the disease classes, they also assign substantial saliency to leaf margins and to the uniform background. We show representative single examples per class, so this is illustrative rather than a quantitative attribution measurement; nonetheless it is visually consistent with the background bias that Noyan [8] and Barbedo [9] documented, suggesting that part of the models’ apparent skill may derive from dataset artefacts rather than from pathology. Pending a quantitative test (e.g. background ablation), the headline accuracies are prudently read as an upper bound on real-world, in-field performance.

5 Discussion

Our results support a two-sided conclusion. On the one hand, a small CNN trained from scratch on a laptop *outperforms*—significantly and reproducibly—the frozen-backbone transfer-learning baselines for this three-class problem, while using 7–34× fewer parameters. This should be read with care: we freeze the pretrained backbones and train only a new head, the standard low-compute transfer recipe, so the comparison is specifically *from-scratch end-to-end training versus frozen-feature transfer under an equal, modest compute budget*. There is also an asymmetry worth stating plainly: because their backbones are frozen, the baselines update only 1,539–3,843 head parameters during training, whereas the from-scratch CNN optimises all 328,587 of its weights. Our “fewer parameters” advantage therefore refers to *total* model size—what must be stored and executed at inference—not the number of weights updated during training. Full fine-tuning of the backbones would very likely raise their accuracy and could narrow or reverse the gap, but it is also substantially more expensive and data-hungry—precisely the cost we set out to avoid. The practical message stands: for a narrow, well-defined diagnostic task on accessible hardware, a compact bespoke model is not a compromise but a strong and far cheaper choice. On the other hand, the Grad-CAM analysis is a cautionary result

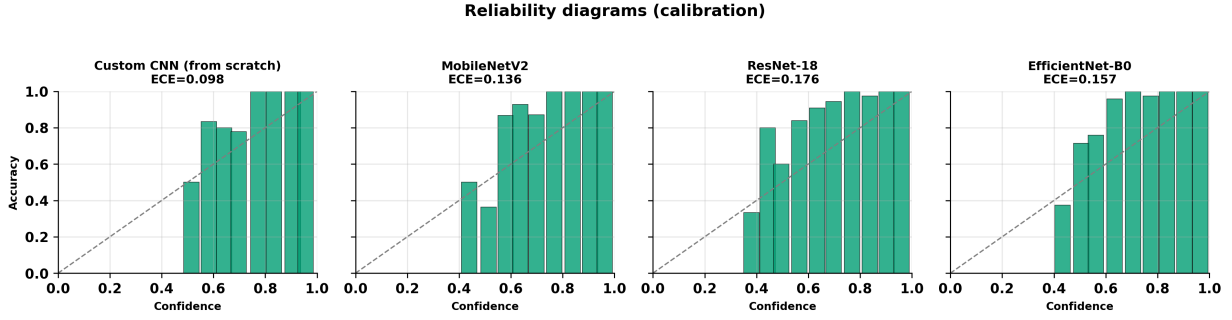


Figure 5: Reliability diagrams (calibration). The dashed diagonal is perfect calibration; bars show accuracy within each confidence bin.

that no amount of additional accuracy on PlantVillage can resolve: because the dataset’s backgrounds are informative, high within-dataset scores do not guarantee field robustness.

Limitations. The study is deliberately scoped. PlantVillage images are laboratory captures with clean backgrounds; field images contain clutter, variable lighting and co-occurring conditions, and we expect all models to degrade on them. The three-class potato task is comparatively easy, which compresses the differences between architectures. We freeze the transfer backbones rather than fully fine-tuning them, trading a little accuracy for speed and a fair, low-compute comparison. We do not deploy on a mobile device, though the parameter counts suggest feasibility.

Future work. The natural next steps are evaluation on field-collected potato images, explicit background-removal or segmentation to force the model onto the leaf, and on-device benchmarking of latency and energy. The reproducible pipeline released here is designed to make those extensions straightforward.

6 Conclusion

We presented a reproducible, statistically careful comparison of a from-scratch CNN against frozen-backbone transfer-learning baselines for potato leaf disease classification on commodity hardware. The compact custom model—more than an order of magnitude smaller than ResNet-18 ($34\times$ fewer parameters)—significantly outperforms the frozen transfer baselines under an equal low-compute budget, and Grad-CAM reveals a shared reliance on dataset background that qualifies all reported accuracies. The work argues for honest, interpretable and hardware-accessible plant-disease modelling over leaderboard chasing.

Author contributions

N.P. conceived and designed the study, implemented the data pipeline, models, training and evaluation code, ran the experiments and statistical analyses, produced the figures, and wrote and revised the manuscript. As a single-author work, all contributions are attributable to N.P.

Data availability

The PlantVillage dataset is publicly available [14]. The download script and the exact potato subset used are described in the code repository; no image data are redistributed here.

Code availability

All code, configurations, pinned dependencies and scripts to reproduce every number and figure in this manuscript are available at <https://github.com/nielspacheco1997/papa-vision>.

Competing interests

The author declares no competing interests.

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Grad-CAM: where each model looks (per class)

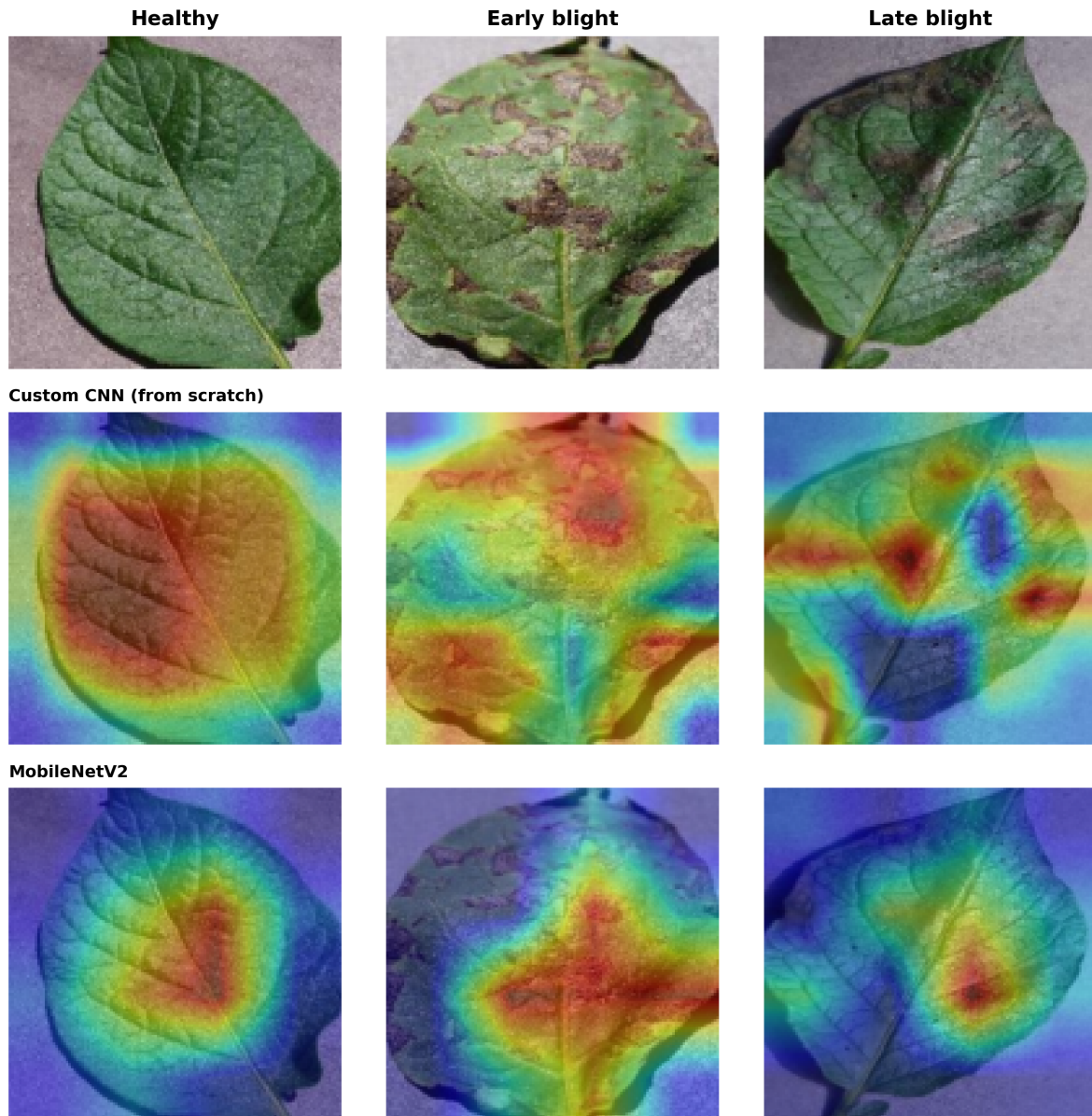


Figure 6: Grad-CAM overlays for one representative example per class. Warm regions are the most influential for the predicted class. Attention spills onto leaf margins and background, illustrating a reliance that is consistent with the PlantVillage background bias (qualitative; not a quantitative attribution metric).